

Example Sediment Transport Problems and Solutions (ENCN442)

1. A trapezoidal canal has a bottom width of 20m side slopes of 1.75H:1V and a slope of 0.0005. The bed and bank material has $d_{75} = 10\text{mm}$.
 - A. Are the banks and bed stable when the canal is flowing at the design depth of 1.4m? Assume an angle of repose of 33° .
 - B. What is the design discharge?

Solution:

Part A: Are the banks and bed stable?

The side slope of the canal is 1.75:1 so, slope $\alpha = \tan^{-1} \left[\frac{1}{1.75} \right] = 29.75^\circ$

Angle of repose = 33°

$k = \left[1 - \frac{\sin^2 \alpha}{\sin^2 \phi} \right]^{1/2} = 0.4125$ which is smaller than 0.75 therefore, the sides of the canal are critical.

The shear stress on the canal bed is calculated as:

$\tau = \rho g h s$ where ρ , g , h and s are the density of water, acceleration due to gravity, depth of flow and slope of the canal. Thus the shear stress on the bed is

$$\tau = 1000 \times 9.81 \times 1.4 \times 0.0005 = 6.867 \text{ Nm}^{-2}$$

The critical shear stress on the bed for coarse material is given by

$\tau_{cb} = 0.056 \times (\rho_s - \rho) \times g \times d$ where ρ_s and d are particle density and particle size of the material.

$$\tau_{cb} = 0.056 \times (2650 - 1000) \times 9.81 \times 0.001 = 9.064 \text{ Nm}^{-2}$$

As the shear stress on the canal bed is smaller than the critical shear stress, the canal bed is stable.

The shear stress on the side slopes (bank) can be calculated as:

$$\tau_s = 0.75 \rho g h s = 0.75 \times 1000 \times 9.81 \times 1.4 \times 0.0005 = 5.1503 \text{ Nm}^{-2}$$

The critical shear stress on the side slopes (bank) is:

$$\tau_{cs} = k \times \tau_{cb} = 0.4125 \times 9.064 = 3.739 \text{ Nm}^{-2}$$

The shear stress on the sides is greater than the critical shear stress on the bed; therefore the sides (bank) of the canal are unstable.

Part B: What is the design discharge?

Cross-sectional area of the canal $A = 20 \times 1.4 + 1.75 \times 1.4 \times 1.4 = 31.43 \text{ m}^2$

Wetted perimeter $P = 20 + 2 \times 1.4 \times \sqrt{(1 + 1.75^2)} = 25.644 \text{ m}$

Hydraulic radius $R = \frac{A}{P} = 1.2256 \text{ m}$

Manning's coefficient $n = 0.0125 \times (d_{75(mm)})^{\frac{1}{6}} = 0.017614$

Flow velocity $v = \frac{1}{n} \times R^{\frac{2}{3}} \times S^{\frac{1}{2}} = 1.454 \text{ ms}^{-1}$

Design discharge $Q = A \times v = 45.698 \text{ m}^3\text{s}^{-1}$

- 2. Design a trapezoidal channel on a slope of 0.0016 for a discharge of 12m³/s. the channel is to be excavated in gravels with $d_{75} = 32\text{mm}$ and angle of repose 33.5°**

Solution:

Manning's coefficient $n = 0.0125 \times (d_{75(mm)})^{\frac{1}{6}} = 0.0214$

Let us **assume** the side of the trapezoidal canal as 2H:1V

$$\alpha = \tan^{-1} \left[\frac{1}{2} \right] = 26.57^\circ$$

Now calculating k, $k = \left[1 - \frac{\sin^2 \alpha}{\sin^2 \phi} \right]^{1/2} = 0.586$ which is less than 0.75 thus sides are critical on stability.

Depth of the flow is $\leq \frac{k \times d}{0.75 \times 11 \times s} = 1.42 \text{ m}$ which is the maximum allowable depth

of flow for that canal side slope, and materials. We will apply a large safety factor and assume our depth should be:

$d = 1.00 \text{ m}$

Assume free board = 0.50 m

Overall depth of the canal = 1.50 m

Area of cross-section $A = b \times 1 + 2 \times 1 \times 1 = (2 + b) \text{ m}^2$

Wetted perimeter $P = (b + 4.47) \text{ m}$

Hydraulic radius $R = \frac{A}{P} = \frac{2+b}{4.47+b}$

Discharge $Q = A x \frac{1}{n} x R^{\frac{2}{3}} x S^{\frac{1}{2}} = (2+b) x \frac{1}{0.0214} x \left[\frac{2+b}{4.47+b} \right]^{\frac{2}{3}} x (0.0016)^{\frac{1}{2}}$

By using a trial and error method of calculation we get $b = 5.75$ m

Next we calculate the stability of the canal bed and sides

Shear stress on bed $\tau = \rho g h s = 1000 x 9.81 x 0.0016 x 1 = 15.696 \text{ Nm}^{-2}$

Critical shear stress on bed

$\tau_{cb} = 0.056 x (\rho_s - \rho) x g x d = 0.056 x (2650 - 1000) x 9.81 x 0.032 = 29.006 \text{ Nm}^{-2}$

The canal bed is stable.

Shear stress on sides: $\tau_{cs} = 0.75 \rho g h s = 0.75 x 1000 x 9.81 x 1 x 0.0016 = 11.772 \text{ Nm}^{-2}$

Critical side shear stress:

$\tau_{cs} = k x \tau_{cb} = 0.586 x 29.006 = 16.996 \text{ Nm}^{-2}$

The canal sides are stable.